

Categories, Proofs, and Processes:

Lecture 18 Monoidal Categories

Wira Azmoon Ahmad

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8 Monoidal Categories

Definition 8.1 (Informal). Monoidal categories are categories with a "parallel" composition $\otimes$.

8.1 Motivating Examples

Example 8.2 ($\mathbf{Set}, \times$).

$$\begin{array}{ccccc}
 f : A \rightarrow B, & g : C \rightarrow D & & & \\
 f \times g : A \times C \rightarrow B \times D & & & & \\
 (f \times g)(a, c) = (f(a), g(c)) & & & & \\
 \begin{array}{ccccc}
 A & \xleftarrow{\pi_1} & A \times C & \xrightarrow{\pi_2} & C \\
 f \downarrow & & \downarrow f \times g & & \downarrow g \\
 B & \xleftarrow{\pi'_1} & B \times D & \xrightarrow{\pi'_2} & D
 \end{array}
 \end{array}$$

Any $\mathcal{C}$ with products

$$f \times g := \langle f \circ \pi_1, g \circ \pi_2 \rangle$$

$$\begin{aligned}
 (f' \times g') \circ (f \times g) &= (f' \times g') \circ \langle f \circ \pi_1, g \circ \pi_2 \rangle \\
 &= \langle f' \circ f \circ \pi_1, g' \circ g \circ \pi_2 \rangle \\
 &= (f' \circ f) \times (g' \circ g) \quad (\text{interchange law})
 \end{aligned} \tag{1}$$

Example 8.3 ($\mathbf{Set}, +$). $+$ is the disjoint union.

$$\begin{aligned}
 f : A \rightarrow B, \quad g : C \rightarrow D \\
 f + g : A + C \rightarrow B + D \\
 (f + g)(a, c) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in C \end{cases}
 \end{aligned}$$

In $\mathcal{C}$ with coproducts:

$$f + g = [\iota_1 \circ f, \iota_2 \circ g]$$

”Co-tupling of maps”

$$\begin{array}{ccccc}
 A & \xrightarrow{\iota_1} & A + C & \xleftarrow{\iota_2} & C \\
 f \downarrow & & \downarrow f+g & & \downarrow g \\
 B & \xrightarrow{\iota'_1} & B + D & \xleftarrow{\iota'_2} & D
 \end{array}$$

$$(f' + g') \circ (f + g) = (f' \circ f) + (g' \circ g)$$

8.2 Non-Examples

Example 8.4 ($\mathbf{Rel}, \times$). $\times$ is not a product or a coproduct since relations are non-separable. Let

$$\begin{aligned}
 \mathbb{B} &= \{0, 1\} \\
 S &= \{ * \mapsto (0, 0), * \mapsto (1, 1) \} \\
 R &= \{ * \mapsto 0, * \mapsto 1 \} \\
 \begin{array}{ccccc}
 \mathbb{B} & \xleftarrow{\pi_1} & \mathbb{B} \times \mathbb{B} & \xrightarrow{\pi_2} & \mathbb{B} \\
 & \swarrow & \uparrow S & \searrow & \\
 & & \{ * \} & &
 \end{array} \\
 S' &= \begin{cases} * \mapsto (0, 0) \\ * \mapsto (0, 1) \\ * \mapsto (1, 0) \\ * \mapsto (1, 1) \end{cases}
 \end{aligned}$$

$S' \neq S$, no unique relation.

Example 8.5 ($\mathbf{Vect}_{\mathbb{C}}, \otimes$). $\otimes$ is the tensor product.

8.3 Definitions

Definition 8.6 (Def. 114). A monoidal category $(\mathcal{C}, \otimes, I)$ is a category $\mathcal{C}$ with a functor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ called the monoidal product, an object $I \in \text{Ob}(\mathcal{C})$ called the unit and natural isomorphisms

$$\begin{aligned}
 \alpha_{A,B,C} : (A \otimes B) \otimes C &\xrightarrow{\sim} A \otimes (B \otimes C) \\
 \lambda_A : I \otimes A &\xrightarrow{\sim}, \quad \rho_A : A \otimes I \xrightarrow{\sim} A
 \end{aligned}$$

satisfying 2 coherence conditions.

Example 8.7.

$$\{(a, b), c | a, b, c\} \cong \{a, (b, c) | a, b, c\}$$

Definition 8.8. A monoidal category is strict if α, λ, ρ are all identities.

Claim 8.9. Any monoidal category is equivalent to a strict category.

8.4 Aside: Term rewrite systems

Definition 8.10. A signature

$$\Sigma = \{\otimes : 2, I : 0\}$$

where $\otimes$ is a symbol, 2, 0 are arities, with arity 0 $\Rightarrow$ constant.

Definition 8.11. A term $t \in T_\Sigma(V)$ for some set V (variables) is a tree whose nodes are in Σ and leaves are in V .

Example 8.12.

$$V = \{x, y, z\}, \quad T_\Sigma V \ni \begin{cases} (x \otimes x) \otimes x \\ I \otimes I \\ (x \otimes y) \otimes z \\ \dots \end{cases}$$

Definition 8.13. A rewrite rule is a pair of terms

$$\begin{aligned} (x \otimes y) \otimes z &\xrightarrow{R_1} x \otimes (y \otimes z) \\ I \otimes x &\xrightarrow{R_2} x \\ x \otimes I &\xrightarrow{R_3} x \end{aligned}$$

Rewrite rules can be applied by

1. substituting variables
2. replacing a sub-term

For

$$\begin{aligned} x &\mapsto a, y \mapsto b, z \mapsto I \\ (a \otimes b) \otimes I &\xrightarrow{R'_1} a \otimes (b \otimes I) \\ ((a \otimes b) \otimes I) \otimes d &\xrightarrow{R'_1 \otimes d} (a \otimes (b \otimes I)) \otimes d \end{aligned}$$

Definition 8.14. "Good" rewrite systems are:

- terminating (all sequences $t \rightarrow \dots \rightarrow t'$ are finite)
- confluent:

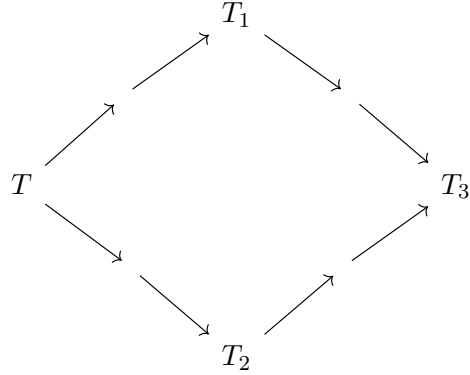
$$\forall t_1, t_2. \left(t \begin{cases} \rightarrow \rightarrow \rightarrow t_1 \\ \rightarrow \rightarrow \rightarrow t_2 \end{cases} \right), \exists t_3. \left(\begin{cases} t_1 \rightarrow \rightarrow \rightarrow \\ t_2 \rightarrow \rightarrow \rightarrow \end{cases} t_3 \right)$$

Terminating + confluent $\Rightarrow$ unique normal forms.

Example 8.15 (Confluence).

$$\begin{array}{ccc} (x \otimes I) \otimes y & \longrightarrow & x \otimes (I \otimes y) \\ \downarrow & \swarrow & \\ x \otimes y & & \end{array}$$

Definition 8.16. "Coherence = confluence + commuting diagrams"



commutes.

Definition 8.17 (MacLane's coherence conditions).

$$\begin{array}{ccc}
 (A \otimes I) \otimes B & & \\
 \alpha \downarrow & \searrow \rho \otimes B & \\
 A \otimes (I \otimes B) & \xrightarrow{A \otimes \lambda} & A \otimes B
 \end{array}$$

$$\begin{array}{ccccc}
 & (A \otimes B) \otimes (C \otimes D) & & & \\
 \alpha \nearrow & & \searrow \alpha_{A \otimes B, C, D} & & \\
 ((A \otimes B) \otimes C) \otimes D & & & & A \otimes (B \otimes (C \otimes D)) \\
 \searrow \alpha \otimes D & & & \nearrow A \otimes \alpha & \\
 & (A \otimes (B \otimes C)) \otimes D \xrightarrow{\alpha} A \otimes ((B \otimes C) \otimes D) & & &
 \end{array}$$

(Δ) (top) and ($\triangleleft$) (bottom) coherence.

Theorem 8.18 (Coherence). *All formal* diagrams of λ, ρ, α commute.*

8.5 Monoidal Categories

Definition 8.19. String diagram notation

$$f : A \rightarrow B$$

$$f : A \otimes B \rightarrow C \otimes D \otimes E$$

$$f \otimes g$$

$$\sigma_{A,B} = \text{swap}_{A,B}$$

Symmetry natural isomorphisms.

Definition 8.20. A symmetric monoidal category (SMC) is a monoidal category with a natural isomorphism

$$\sigma_{A,B} A \otimes B \xrightarrow{\sim} B \otimes A$$

satisfying:

$$\begin{array}{ccccc} & & \sigma_{A,B}^{-1} = \sigma_{B,A} & & \\ & & \downarrow & & \\ & A \otimes (B \otimes C) & \xrightarrow{\sigma_{A,B \otimes C}} & (B \otimes C) \otimes A & \\ \alpha \nearrow & & & & \searrow \alpha \\ (A \otimes B) \otimes C & & & & B \otimes (C \otimes A) \\ \searrow \sigma_{A,B \otimes 1_C} & & & & \nearrow \sigma_{A,C} \\ & (B \otimes A) \otimes C & \xrightarrow{\alpha} & B \otimes (A \otimes C) & \end{array}$$

Note 8.21. Braided monoidal category makes distinction of over or under swap.

Definition 8.22. A strict SMC is an SMC where $\alpha = 1, \lambda = \rho$. ($\sigma \neq 1!$).

8.6 Algebric structures in monoidal categories

Definition 8.23. A monoid in a strict monoidal category $(\mathcal{C}, \otimes, I)$ is an object X with morphisms

$$\mu : X \otimes X \rightarrow X, \quad \eta : I \rightarrow X$$

satisfying:

- associativity

$$\begin{array}{ccc} X \otimes X \otimes X & \xrightarrow{X \otimes \mu} & X \otimes X \\ \mu \otimes X \downarrow & & \downarrow \mu \\ X \otimes X & \xrightarrow{\mu} & X \end{array}$$

- left unit

$$\begin{array}{ccc} X & \xrightarrow{\eta \otimes X} & X \otimes X \\ & \searrow 1_X & \downarrow \mu \\ & & X \end{array}$$

- right unit

$$\begin{array}{ccc} X & \xrightarrow{X \otimes \eta} & X \otimes X \\ & \searrow 1_X & \downarrow \mu \\ & & X \end{array}$$

Note 8.24.

$$\begin{aligned} & (- \otimes X) : \mathcal{C} \rightarrow \mathcal{C} \\ f \otimes X &:= f \otimes 1_X : A \otimes X \rightarrow B \otimes X \end{aligned}$$

Example 8.25. Monoids in $(\mathbf{Set}, \times, \{*\})$ are monoids.

$$(X, \mu : X \times X \rightarrow X, \eta : \{*\} \rightarrow X)$$

$$a \cdot b := \mu(a, b)$$

$$1 := \eta(*)$$

- Associativity says

$$\forall a, b, c \in X, \mu(a, \mu(b, c)) = \mu(\mu(a, b), c)$$

$$\text{i.e. } a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

- Left unit

$$\forall a \in X, \mu(\eta(*), a) = a$$

$$\text{i.e. } (1 \cdot a) = a$$

- Right unit

$$\forall a \in X, a \cdot 1 = a$$

Example 8.26. Monoids in $(\mathbf{Ab}, \times, \{0\})$ - where $\mathbf{Ab}$ is the category of abelian groups and group homomorphisms - are rings.

$$(X, \mu : X \times X \rightarrow X, \eta : \{0\} \rightarrow X)$$

as before, but X is an abelian group and μ and η must be group homomorphisms.

$$\mu(a + b, c) = \mu(a, c) + \mu(b, c)$$

$$\mu(a, b + c) = \mu(a, b) + \mu(a, c)$$

i.e.

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

Example 8.27. Monoids in $(\mathbf{Vect}_k, \otimes, k)$ are associative algebras, $:=$ vector space with multiplication $(- \cdot -)$ which is associative, unital and linear in each argument.

$$\left(\sum_i \lambda_i v_i \right) \cdot w = \sum_i \lambda_i (v_i \cdot w)$$

$$v \cdot \left(\sum_i \lambda_i w_i \right) = \sum_i \lambda_i (v \cdot w_i)$$